CAM – MIDTERM EXAM 06/11/2020

SOLUTIONS TO NUMERICAL PROBLEMS

METAL FORMING

A cylindrical sample in 6061-O aluminum alloy (K = 205 MPa, n = 0,2) with an initial height $h_0 = 50$ mm and an initial diameter $d_0 = 40$ mm, is compressed between two flat dies.

a. Determine the diameter of the sample if the true strain is equal to $\varepsilon = -0{,}16$.
b. The process requires two steps: a first one to a height $h_1 = 43$ mm, followed by a second step to a height $h_2 = 37$ mm. Neglecting friction, determine the force required for each step, and the ideal work spent on the overall deformation.
c. In the hypothesis that the strain energy density to reach the final shape is equal to $u = 0{,}15$ J/mm^3, determine the cost of the energy required to deform 5000 samples (cost of the energy equal to $0{,}1$ €/kWh).

Solution

a. Determine the diameter of the sample if the true strain is equal to $\varepsilon = -0{,}16$.

Considering the volume as constant:

$$V_0 = \frac{\pi d_0^2}{4} h_0 = V_f = \frac{\pi d_f^2}{4} h_f$$

Being the final height equal to:

$$\varepsilon = \ln\left(\frac{h_f}{h_0}\right)$$

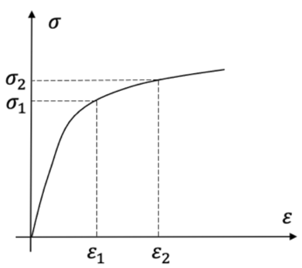
$$h_f = h_0 e^{\varepsilon} = 50 \cdot e^{-0{,}16} = 42{,}61 \text{ mm}$$

The final diameter of the sample is equal to:

$$d_f = \sqrt{\frac{h_0}{h_f} d_0^2} = \sqrt{\frac{50}{42{,}61} 40^2} = 43{,}33 \text{ mm}$$

b. The process requires two steps: a first one to a height $h_1 = 43$ mm, followed by a second step to a height $h_2 = 37$ mm. Neglecting friction, determine the force required for each step, and the ideal work spent on the overall deformation.

Referring to the figure, the required forces are:



$$F_1 = \sigma_1 A_1$$

$$F_2 = \sigma_2 A_2$$

Where, the flow stresses are:

$$\sigma_1 = k\varepsilon_1^n$$

$$\sigma_2 = k\varepsilon_2^n$$

Being:

$$\varepsilon_1 = \ln\left(\frac{h_1}{h_0}\right) = \ln\left(\frac{43}{50}\right) = -0{,}15$$

$$\varepsilon_2 = \ln\left(\frac{h_2}{h_0}\right) = \ln\left(\frac{37}{50}\right) = -0{,}30$$

$$\sigma_1 = k\varepsilon_1^n = 205 \cdot (-0{,}15)^{0{,}2} = -140{,}27 \text{ MPa}$$

$$\sigma_2 = k\varepsilon_2^n = 205 \cdot (-0{,}30)^{0{,}2} = -161{,}13 \text{ MPa}$$

and the cross section of the specimen in the two steps equal to:

$$A_1 = \frac{V_0}{h_1} = \frac{\frac{\pi d_0^2}{4} h_0}{h_1} = \frac{\frac{\pi 40^2}{4} 50}{43} = 1.461 \text{ mm}^2$$

$$A_2 = \frac{V_0}{h_2} = \frac{\frac{\pi d_0^2}{4} h_0}{h_2} = \frac{\frac{\pi 40^2}{4} 50}{37} = 1.698 \text{ mm}^2$$

The required forces are:

$$F_1 = \sigma_1 A_1 = -140{,}27 \cdot 1.461 = -204.934 \text{ N} = -204{,}9 \text{ kN}$$

$$F_2 = \sigma_2 A_2 = -161{,}13 \cdot 1.698 = -273.599 \text{ N} = -273{,}6 \text{ kN}$$

The overall strain energy density is equal to:

$$u = \left|\frac{k\varepsilon_2^{n+1}}{n+1}\right| = \left|\frac{205 \cdot (-0{,}30)^{0{,}2+1}}{0{,}2+1}\right| = 40{,}3 \text{ MPa} = 0{,}0403 \ \frac{\text{J}}{\text{mm}^3}$$

Therefore, the ideal work spent on the overall deformation is equal to:

$$W = uV_0 = 0{,}0403 \cdot \frac{\pi 40^2}{4} 50 = 2.532 \text{ J} = 2{,}5 \text{ kJ}$$

c. *In the hypothesis that the strain energy density to reach the final shape is equal to* $u = 0{,}15$ *J/mm³, determine the cost of the energy required to deform 5000 samples (cost of the energy equal to* $0{,}1$ *€/kWh).*

The ideal work to produce a single sample is equal to:

$$W = uV_0 = 0{,}15 \cdot \frac{\pi 40^2}{4} 50 = 9.425 \text{ J}$$

The ideal work to produce 5.000 samples is equal to (recall $1 \text{ kWh} = 3{,}6 \text{ MJ}$):

$$W_{total} = nW = 5.000 \cdot 9.425 = 47.125.000 \text{ J} = 47{,}125 \text{ MJ} = \frac{47{,}125}{3{,}6} \text{kWh} = 13{,}1 \text{ kWh}$$

Therefore, the cost of energy is equal to:

$$C_E = W_{total} \cdot c_{ele} = 13{,}1 \text{ kWh} \cdot 0{,}1 \frac{€}{\text{kWh}} = 1{,}31 \text{ €}$$

METAL CASTING

The Figure 1 shows a casting to be fabricated by an expendable mould process. The casting will be made of AISI 304 steel (for simplicity, draft angles and fillets/radii are omitted). The casting is divided in 5 elementary geometries (numbered from 1 to 5).

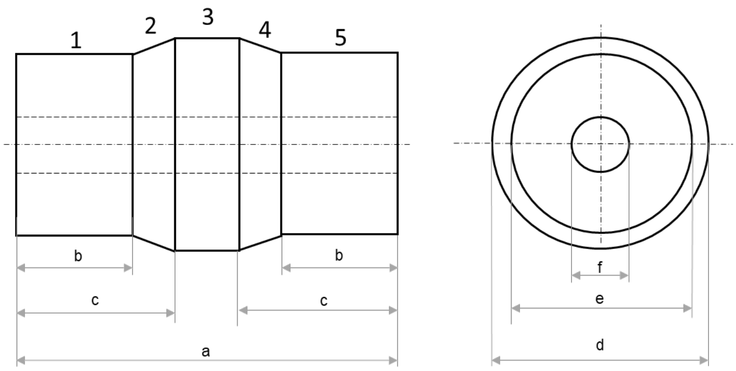


Figure 1

a. Please, study the directional solidification of the metal. Dimensional data: a = 340 mm, b = 115 mm, c = 145 mm, d = 165 mm, e = 145 mm, f = 60 mm.
b. Considering: the casting axis as parting line, A_s:A_r:A_g = 2:2:1, c = 0,6, two gates with an area equal to 700 mm^2 each, a volume of the casting equal to 5 dm^3, a top riser with a volume equal to 3 dm^3, a cope height equal to 300 mm. Design a middle gating system, verifying if the mould filling time is lower than 10 s.

Solution

a. Please, study the directional solidification of the metal.

In order to study the directional solidification of the metal, we need to evaluate the M=V/A ratio for each of the elementary geometries.

Geometry n.1 (equal to Geometry n.5):

$$V_1 = \frac{\pi}{4}(e^2 - f^2)b = \frac{\pi}{4}(145^2 - 60^2)115 = 1.573.840 \text{ mm}^3$$

$$A_1 = \pi(e + f)b + \frac{\pi}{4}(e^2 - f^2) = \pi(145 + 60)115 + \frac{\pi}{4}(145^2 - 60^2) = 87.749 \text{ mm}^2$$

$$M_1 = \frac{V_1}{A_1} = \frac{1.573.840}{87.749} = 17{,}94 \text{ mm} = M_5$$

Geometry n.2 (equal to Geometry n.4):

Being $L_2 = c - b = 30$ mm the length of the truncated cone,

$$V_2 = \frac{1}{12}\pi(d^2 + e^2 + de)L_2 - \frac{\pi}{4}(f^2)L_2 = \frac{1}{12}\pi(165^2 + 145^2 + 165 \cdot 145)30 - \frac{\pi}{4}(60^2)30 = 482.038 \text{ mm}^3$$

$$A_2 = \frac{\pi}{2}(d + e)\sqrt{{L_2}^2 + \left(\frac{d}{2} - \frac{e}{2}\right)^2} + \pi f L_2 = \frac{\pi}{2}(165 + 145)\sqrt{30^2 + \left(\frac{165}{2} - \frac{145}{2}\right)^2} + \pi 60 \cdot 30 = 21.054 \text{ mm}^2$$

$$M_2 = \frac{V_2}{A_2} = \frac{482.038}{21.054} = 22{,}90 \text{ mm} = M_4$$

Geometry n.3:

Being $L_3 = a - 2c = 50 \text{ mm}$ the length of cylinder,

$$V_3 = L_3 \frac{\pi}{4}(d^2 - f^2) = 50\frac{\pi}{4}(165^2 - 60^2) = 927.752 \text{ mm}^3$$

$$A_3 = \pi(d + f)L_3 = \pi(165 + 60)50 = 35.343 \text{ mm}^2$$

$$M_3 = \frac{V_3}{A_3} = \frac{927.752}{35.343} = 26{,}25 \text{ mm}$$

So $M_3 > M_2 > M_1$ and $M_3 > M_4 > M_5$, with

$$\frac{M_2}{M_1} = \frac{M_4}{M_5} = \frac{22{,}90}{17{,}94} = 1{,}28 \in [1{,}1 \div 1{,}3]$$

$$\frac{M_3}{M_2} = \frac{M_3}{M_4} = \frac{26{,}25}{22{,}90} = 1{,}15 \in [1{,}1 \div 1{,}3]$$

Therefore, the solidification will start in geometries 1 and 5, and will end in geometry 3.

A riser should be positioned in contact with the geometry 3.

b. Design a middle gating system, verifying if the mould filling time is lower than 10 s.

In order to find the mould filling time, we need to know the area of the bottleneck and the velocity of the molten metal in this area. Being by design $A_s : A_r : A_g = 2 : 2 : 1$, the bottleneck is in the gates. The overall area of the gates is equal to:

$$A_g = 2 \cdot 700 = 1.400 \text{ mm}^2$$

In order to determine the velocity, being a middle gating system, we need to recall that:

$$v = c\sqrt{2gH_{av}}$$

where:

$$H_{av} = \left(\frac{1}{\frac{r_{drag}}{\sqrt{h_i}} + \frac{r_{cope}}{\frac{\sqrt{h_i} + \sqrt{h_f}}{2}}} \right)^2$$

$$V_{total} = V_{casting} + V_{riser} = 5 + 3 = 8 \text{ dm}^3$$

$$V_{cope} = \frac{V_{casting}}{2} + V_{riser} = 2{,}5 + 3 = 5{,}5 \text{ dm}^3 \rightarrow r_{cope} = \frac{V_{cope}}{V_{total}} = \frac{5{,}5}{8} = 0{,}6875 = 68{,}75\%$$

$$V_{drag} = \frac{V_{casting}}{2} = 2{,}5 \text{ dm}^3 \rightarrow r_{drag} = \frac{V_{drag}}{V_{total}} = \frac{2{,}5}{8} = 0{,}3125 = 31{,}25\%$$

Being $h_i = 300 \text{ mm}$ and $h_f = 0 \text{ mm}$ (we are using a top riser), we get:

$$H_{av} = \left(\frac{1}{\frac{0,3125}{\sqrt{300}} + \frac{0,6875}{\frac{\sqrt{300}+\sqrt{0}}{2}}} \right)^2 = 105,35 \text{ mm} = 0,10535 \text{ m}$$

Therefore:

$$v = c\sqrt{2gH_m} = 0,6\sqrt{2 \cdot 9,81 \cdot 0,10535} = 0,863\frac{\text{m}}{\text{s}} = 863\frac{\text{mm}}{\text{s}}$$

and the mould filling time is:

$$T_{MF} = \frac{V_{total}}{A_g \cdot v} = \frac{8.000.000 \text{ mm}^3}{1.400 \text{ mm}^2 \cdot 863\frac{\text{mm}}{\text{s}}} = 6,62 \text{ s } < 10 \text{ s}$$

The mould filling time satisfies the constraint.